

WRRRA Prime Parts Rendering Followup 0.1–0.12: Reviewed and Corrected Collection

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This prime-parts series is distinct from the WRRRA_M upstream/downstream synthesis.

Abstract

The original 0.1–0.10 calculations were run in isolated directories, and the 0.11 shared-origin/denominator/energy ledger and 0.12 finite-pool first-generation spatial feedback were audited again. Ten stage results agree numerically with their stored outputs. Stage 0.11 passes 47 cross-stage checks and 26 separate audits; 0.12 passes 44 internal and 12 separate audits. Stages 0.1 and 0.2 agree numerically but differ in JSON serialization bytes; 0.3–0.10 outputs also match byte-for-byte. These tests establish implementation reproducibility, not ten independent observational confirmations. Under the declared spatial configuration the first-generation fusion-photon absorption expectation is 134.812677. No further stage number is a release condition for this scoped collection.

Assessment: verified inputs WRRRA transformation outputs falsifiers

Verified inputs: finite arithmetic $N=1,000,000$; prime component labels 5 and 47; recorded $\alpha=1.8996876950554356$ and $\beta=0.8654570124136961$; measured nucleon, electron and nuclear rest energies; neutron mean lifetime 878.3 s; the K15 small-first pool; and explicitly supplied radiation, contact, cooling and geometry choices. Calibration and configuration choices are recorded as model inputs.

WRRRA transformation: finite addresses/components Q/B/L and shared source label-preserving internal transitions capture/fusion/beta competition binding-energy photon and recoil ledger conditional target states and spatial contacts first-generation fusion-photon feedback in the declared common pool. An arithmetic address share, per-address energy, per-pool energy and six-baryon conditional probability have different denominators and are never directly summed.

Outputs: arithmetic weighted admitted-address share 5%; shared conditional address $B=15.982097259128$, $E=15319.262597108$ MeV; K15 common pool $B=1,000,000$, active $B=999,219$, source $E=958,526,427.960422$ MeV. At $t=180$ s with chosen hazard ratios, fusion 63,343.786871, capture 127,954.455314, new beta 1,279.544553 and total pool photons 1,790,199.616097 MeV. The separately prepared six-baryon ray has conditional absorption probability 10.640952% and surviving target deuteron probability 99.886854%. The declared pool-wide first-generation ray construction yields 134.812677 absorptions, 133.477884 recaptures and 1,265.546779 remaining deuterons.

Falsifiers: duplicated prime/component/photon source consumption; violation of source energy, Q/B/L, baryon, four-momentum or probability conservation; absorption under $q=0$; or disagreement with conditional observations under fixed geometry and inputs. A later change of configuration must be documented and retested as a new instance.

1. Stage genealogy and review

Stage 0.1 separated admitted-address ratios, units and denominators. Stage 0.2 tested alternative supply assignments using measured masses; those energy-only candidates cannot be merged with the common B/E envelope of 0.4. Stage 0.3 distinguished n from $p+e+\text{antineutrino}$ readouts under a source with $L=0$ and explicit

Q/B/L constraints. Stage 0.4 implemented the same B/E source envelope, protected reserve, prime-label-preserving current and lifetime-calibrated transition across four assemblies. Stage 0.5 computed finite pairing, K15 versus K16 ordering, deuteron/helium binding and beta competition. The K16 baseline helium percentage must not be identified with the selected K15 pool percentage.

Stage 0.6 closed capture photon 2.223248712 MeV and fusion photon 23.770682375 MeV plus recoil against their binding differences. Stage 0.7 tested cooling/readiness and reverse thresholds. Stage 0.8 coupled flight, absorption and recapture of a source quartet photon and a separate target in a selected six-baryon preparation. Stage 0.9 supplied six directional axes, finite contact attempts and grid placement; prime identity was not interpreted as position. Stage 0.10 audited 23 chosen event paths for four-momentum across nuclei, photons, environment and external work. Stage 0.11 linked the 0.4–0.6 common-pool source and the distinct 0.7–0.10 selected event by provenance, retaining separate denominators. Stage 0.12 extended to expected first-generation photon/target responses across the configured pool.

2. Actual spatial calculation and correction

Freeze the 0.6 source population at $t=180$ s. Place $Q=64,617$ quartet sites in a line ten light-seconds apart. A photon emitted in the $+x$ direction (one of six equally weighted axes) from site i may reach deuteron target $i+1$; the last site has no next target. Assign independent neighboring fusion and deuteron marginals F/Q and D/Q . Perform ten Bernoulli contact attempts with $q=.1$. Thus the expected absorbed first-generation fusion photons are $F \times (Q-1)/Q \times (1/6) \times (D/Q) \times [1-(1-q)^{10}] = 134.812677492537$, where $D=C-2F=1,266.881572721$. The photon pulse has ten seconds of flight and a finite contact window after the frozen source snapshot; target cooling, beta and recapture then evolve for 180 s. Simultaneous source production and radiation transport are not claimed.

The review caught an energy bookkeeping mistake during construction of 0.12: adding ``initial_beta_budget_MeV`` to the already recorded electron plus antineutrino energy would count 680,618.461596 MeV twice. The corrected code asserts absolute closure to the inherited source, 958,526,427.960422 MeV, before and after target feedback. Around 134.81 fusion photons are absorbed and around 133.48 new capture photons are retained; the photon pool changes from 1,790,199.616097 to 1,787,291.781293 MeV with total energy conserved. This is a correction of this release's construction audit, not a retrospective change to measured data.

3. Scope and reproducibility

Neighboring site independence, a pulse after the $t=180$ s source snapshot, ten-light-second geometry, q and cooling are stated choices. Prime labels mark provenance, not coordinates. Photons move at the fixed light speed; nuclei are not transported ten light-seconds. Secondary capture photons remain in the energy ledger but are not routed to another neighbor. Microscopic cross sections, correlated site microstates, simultaneous spacetime evolution and multi-generation absorption have not been computed in this release.

Reproducing known physical inputs is a reality-consistency and explanatory result for WRRRA. The structural originality lies in its executed finite component transformations and the common-source to nuclear and radiative connections. An unmeasured output calculated after inputs have been fixed is a conditional WRRRA prediction. This 0.12 collection is complete for its disclosed executed scope and does not require an indefinitely expanding version number. We do not claim that the arithmetic 5% is directly the physical cosmic-energy fraction, nor that the microscopic nuclear or electromagnetic forces were derived by this component model alone.

The reproduction archive retains original 0.1–0.6 packages, executed 0.7–0.10 sources/results, 0.11 cross-stage code, corrected 0.12 code, replay evidence and SHA-256 manifests. Counts of checks are mathematical, implementation and conservation audits. Historical snapshots are preserved; corrections and interpretation belong to this review edition.